

Housing Price Trends

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Problem Statement:

The purpose of this project is to discuss the trends in the prices of houses in various regions within the United States. The price of houses plays a major factor in affordability, cost of living, and the economy as a whole. The price of houses has been subject to various dynamics in the past few years due to various reasons such as inflation, interest rates, and demand and supply.

Source of Data:

This project will use housing price data [Zillow Home Value Index \(ZHVI\) dataset](#), which provides monthly home value estimates for different metro areas in the United States.

The dataset includes information such as region, state, and home values over time. I selected this dataset because housing prices are an important economic issue that impacts affordability and cost of living. The data was organized so it can be easily analyzed, and a few selected metro areas will be used to compare trends. The analysis will focus on how housing prices have changed over time and how they differ between regions. Power BI will be used to create visualizations such as line charts and comparisons to clearly show trends and patterns in the housing market.

Data Wrangling:

The first row was used as headers, and columns that were not necessary for the analysis were deleted. Since the dataset initially had columns for each date, the data was transformed by unpivoting the date columns to a singular Date column with a corresponding HomeValue column. This resulted in a dataset with a time-series format,

making it easier for analysis. The names of the columns were adjusted accordingly, and the data types were modified. Date was now in the date format, and HomeValue was a decimal. Missing values were checked as well.

Dataset for Power BI:

Sample size:

The dataset consists of multiple monthly housing prices of metro areas in the United States spanning several years. The Date dimension supports time-series analysis, while RegionName and StateName support geographic comparisons.

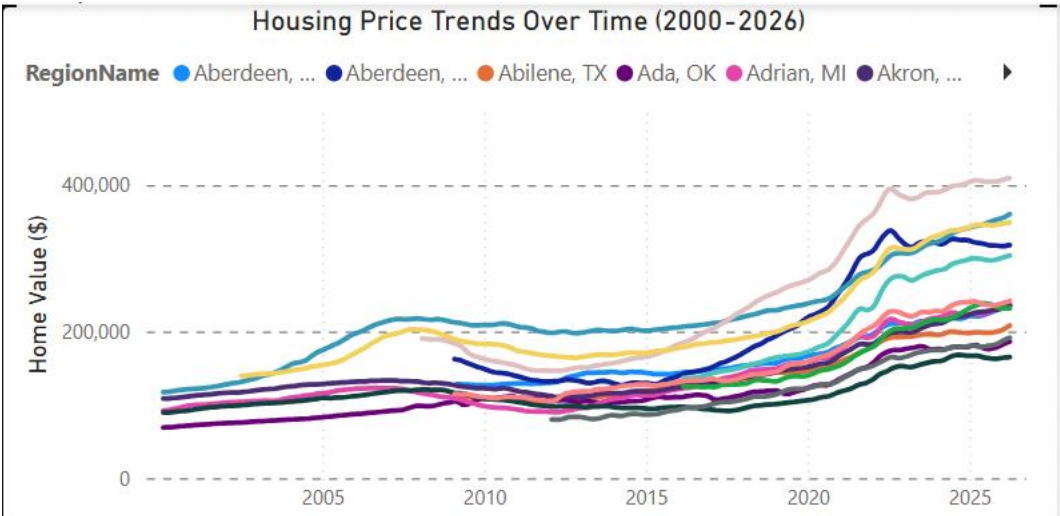
Facts (Measures):

HomeValue

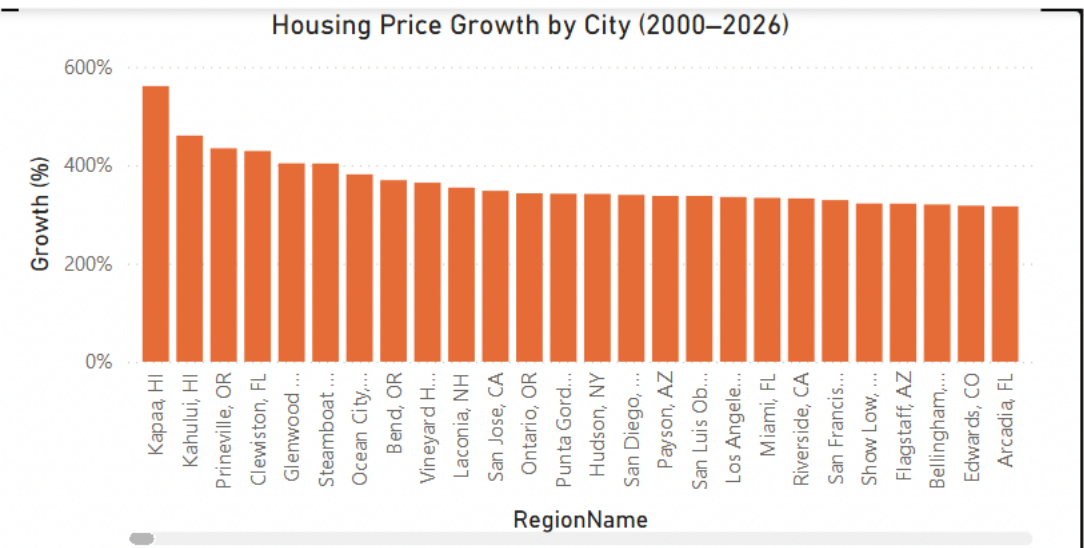
Dimensions:

RegionName, StateName, Date

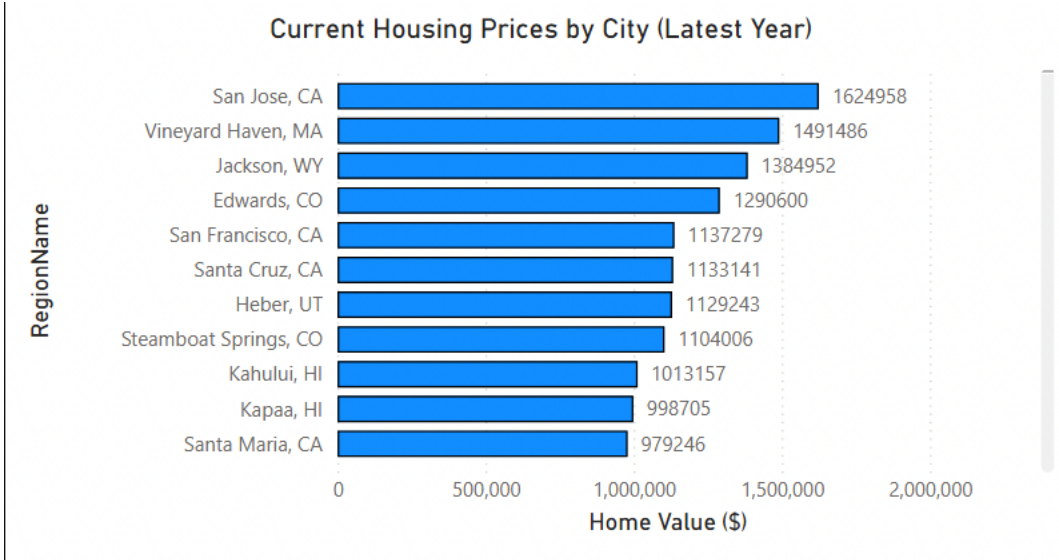
Visualizations:



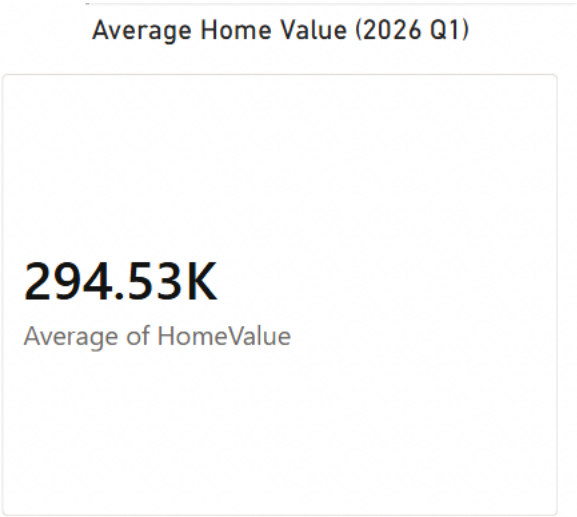
This line chart shows the house prices overtime of individual cities from 2000-2026. We can see that the general trend is going upward.



The clustered column chart compares the percentage growth housing prices from 2000-2026 of metro areas. Some cities show a faster rate of increase compared to others.

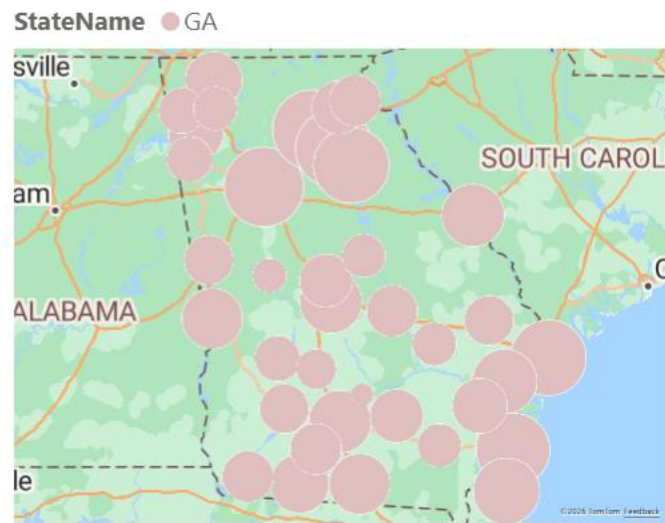


The horizontal bar chart compares the latest housing prices (2026). You can identify the most expensive cities to live in.



This card displays the average home value across metro cities in the first quarter of 2026.

Geographic Distribution of Housing Prices (2026 Q1)



This map visualizes housing prices geographically across metro areas and highlights regional differences in housing markets across the United States (2026 Q1). The map is currently filtered for GA.

Filter by State and City

> ☐ AK

> ☐ AL

> ☐ AR

> ☐ AZ

> ☐ CA

> ☐ CO

> ☐ CT

> ☐ DE

This slicer allows users to filter by state and/or city.

KPI Investigated:

The main KPI investigated was the house prices overtime.

Dashboard, Charts, and Interactivity Tools:

The dashboard was created in Power BI to visualize housing price trends and regional comparisons across metro areas in the United States. The line chart shows the house prices overtime of individual cities from 2000-2026, the clustered column chart compares the percentage growth housing prices from 2000-2026, The horizontal bar chart compares the latest housing prices (2026 Q1), the card displays the average home value across metro cities in the first quarter of 2026, and the map visualizes housing prices geographically across metro areas and highlights regional differences in housing markets across the United States (2026 Q1). A slicer is also added for interaction, allowing users to filter specific cities and/or states for further analysis.

Business/User Value:

This project will be very helpful in gaining valuable insights into the trends followed by the prices of houses over time. It will also help individuals, investors, and the government take the right decisions by considering the trends followed by the prices. The outcome of this project is to gain valuable insights into the trends followed by the prices, which might include the periods of rapid growth, the differences in the prices followed by different regions, and the trends followed by the housing market.

Conclusion and Future Directions:

Using Power BI and the Zillow dataset, the analysis revealed that house prices did indeed increased overtime from 2000 to 2026. Some regions showed larger increases in house prices, while others showed only minimal increases. I also noticed a decline around 2008 to 2012. Upon further research, it was due to the financial crisis of that time. One limitation of the project is that the dataset is that the dataset only showed prices at the metro level. There wasn't any further detailed information such as the counties or specific housing types. In the future, the project could be expanded by including additional factors such as interest rates, inflation, or income levels to further analyze housing affordability and market trends.